EffiSkill: Agent Skill Based Automated Code Efficiency Optimization

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HarnessRSI 67 ★★

evidence: retained

Paper

1 What is built, what is reused

STAGE I · offline, run once

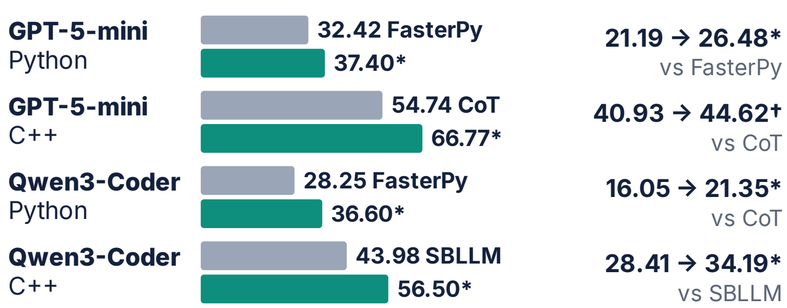
1. **Slow / fast program pairs**
   900 Python + 900 C++ pairs, ratio ≥ 2
2. **Trace → signature → cluster**
   gpt-5.1 traces; TF-IDF + MiniLM, KMeans
3. **Distill and merge skill cards**
   merge at TF-IDF cosine 0.8, then LLM merge

Skill library: 29 Operator Skills + Meta Skills
fixed after Stage I · weights never change

STAGE II · per unseen program

1. **Diagnose**
   bottleneck brief from statement + code
2. **Retrieve**
   3 candidate sets of Operator Skills
3. **Compose plans**
   Meta Skills write 2–3 plans per set
4. **Generate k = 8 candidates**
   never run here; public-test filter, rank

2 Headline: best in all 4 settings



Bars: OPT@8 (%). EffiBench-X, 623 tasks, k = 8, Table 1. * p<0.05, † p<0.10: paired bootstrap over tasks; no repeated runs reported.

3 Ablation: each removal hurts

OPT@1 / OPT@8 (Python)	GPT-5-mini	Qwen3-Coder
EffiSkill (full)	**26.48 / 37.40**	**21.35 / 36.60**
w/o Retrieval	12.20 / 27.45	7.22 / 9.15
w/ Random Skills	13.80 / 27.82	12.36 / 13.80
w/o Multi Plans	18.78 / 25.84	11.88 / 14.29
Instruction (plain prompt)	18.62 / 31.62	12.52 / 24.40
Best baseline	21.19 / 32.42	16.05 / 28.25

Underlined OPT@8: all six are below plain Instruction. Table 2 has no language label; its full rows equal the Python cells. One value per cell, no SD. No w/o-diagnosis, Meta-Skill or non-mined-skills row. Best baseline: FasterPy; Qwen @1 is CoT.

4 What the comparison does not control

- **Budget:** same k = 8 candidates; no tokens, LLM calls or mining cost reported
- **Baselines:** none described as getting the mined library; RAG / FasterPy corpora not stated; SBLLM is a single-iteration variant
- **Held-out:** overlap check stated, method not described; mining and test are both competitive programming
- **C++:** 900 C++ pairs are in the mining corpus, so not zero-shot transfer; pooled vs per-language library is not stated

5 Where the gain is small or unclear

- **OPT@1 gaps only 3.69–5.78‡;** GPT-5-mini C++ OPT@1 is only p<0.10 (about 23 tasks‡)
- OPT@8 gap: **+4.98 / +8.35** Python vs **+12.03 / +12.52** C++; 62.60‡ / 63.40‡ % of Python tasks still unsolved at k = 8
- Fig. 5 "strongest non-EffiSkill baseline" is SBLLM / EffiCoder / Instruction in 3 of 4 panels, not the Table 1 leaders
- Intro / RQ3 box: Qwen gaps 12.36 / 8.67; Table 1: 12.52 / 8.35 (we use the table)

Takeaway

Mined, retrieved skills give the best point estimates in all 4 settings, most on C++ at k = 8 (one OPT@1 cell is only p<0.10). Retrieval and multi-plan composition both matter in the ablations. No run-to-run variance or inference cost is reported, and OPT@1 gains are 3.69–5.78‡.

Our read · HarnessRSI

Score 67 · ★★ · evidence level: retained

Reports: held-out tasks + overlap check, ablations (Table 2), task bootstrap, equal k.

Does not report: seeds / variance, token or mining cost, w/o-diagnosis control, backbone-contamination check.